

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			To get electricity / acts as a back-up source (1) When it is not sunny (or during the night) or when the panels don't generate enough electricity (1) Alternative: Electricity can be fed back [into the National Grid] (1) If it is very sunny or if the panels produce more electricity than is needed (1) Accept energy for electricity / sunlight is not reliable Don't accept solar panels aren't reliable		2		2		
	(c)	(i)		$20 \times 60 = 1\,200$ [A]		1		1	1	
		(ii)	I	Substitution: $230 \times 1\,200$ (ecf) (1) $= 276\,000$ [W] (1) Alternative: Substitution: 230×60 (1) $13\,800 \times 20 = 276\,000$ [W] (1)	1	1		2	2	
			II	$\frac{276\,000 \text{ ecf}}{1000} = 276$ [kW]		1		1	1	
				Question 5 total	7	5	0	12	4	0